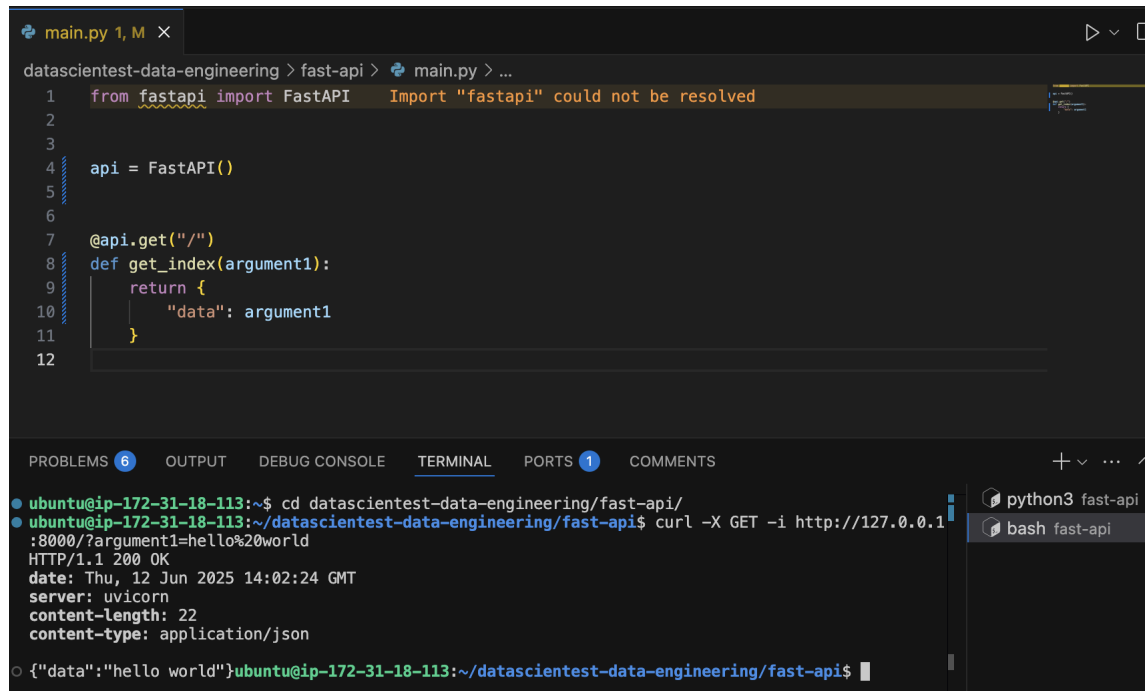


Part b Verification of FastAPI - Pass the data to the API (Request String)

Ensure command below **uvicorn main:api --reload** has run:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ lsof -i :8000
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ uvicorn main:api --reload
INFO: Will watch for changes in these directories: ['/home/ubuntu/datascientest-data-engineering/fast-api']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [1980] using StatReload
INFO: Started server process [1982]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Modify [main.py](#) file and run command below:

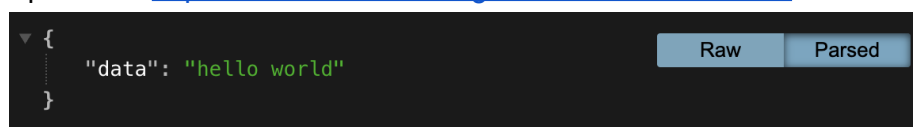


The screenshot shows the VS Code editor with a file named `main.py` open. The code defines a FastAPI application with a single endpoint `get_index` that returns a JSON object with a `data` field. The terminal at the bottom shows the command `curl -X GET -i http://127.0.0.1:8000/?argument1=hello%20world` being executed, and the output is a 200 OK response with a JSON body `{ "data": "hello world" }`.

```
main.py 1, M X
datascientest-data-engineering > fast-api > main.py > ...
1 from fastapi import FastAPI Import "fastapi" could not be resolved
2
3
4 api = FastAPI()
5
6
7 @api.get("/")
8 def get_index(argument1):
9     return {
10         "data": argument1
11     }
12

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS 1 COMMENTS
ubuntu@ip-172-31-18-113:~$ cd datascientest-data-engineering/fast-api/
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i http://127.0.0.1:8000/?argument1=hello%20world
HTTP/1.1 200 OK
date: Thu, 12 Jun 2025 14:02:24 GMT
server: uvicorn
content-length: 22
content-type: application/json
{"data":"hello world"}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$
```

Open URL <http://127.0.0.1:8000/?argument1=hello%20world>



The screenshot shows a JSON viewer displaying the response body `{ "data": "hello world" }`. The viewer has tabs for "Raw" and "Parsed", and the "Parsed" tab is selected.

```
{
  "data": "hello world"
}
```

Run command below:

```
{"data":"hello world"}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i http://127.0.0.1:8000/
HTTP/1.1 422 Unprocessable Entity
date: Thu, 12 Jun 2025 14:03:52 GMT
server: uvicorn
content-length: 95
content-type: application/json

{"detail":[{"type":"missing","loc":["query","argument1"],"msg":"Field required","input":null}]}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$
```

Open OpenAPI gui <http://127.0.0.1:8000/docs>

FastAPI 0.1.0 OAS 3.1

/openapi.json

default ^

GET / Get Index v

Schemas ^

HTTPValidationError > Expand all object

ValidationError > Expand all object

Query for example **Hello World:**

default ^

GET / Get Index ^

Parameters Cancel

Name	Description
argument1 ★ required any (query)	Hello World

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/?argument1=Hello%20World' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/?argument1=Hello%20World
```

Server response

Code	Details
200	Response body <pre>{ "data": "Hello World" }</pre> Download Response headers <pre>content-length: 22 content-type: application/json date: Thu, 12 Jun 2025 14:08:30 GMT server: uvicorn</pre>

Update [main.py](#) file:

```
main.py 1, M X
datascientest-data-engineering > fast-api >
1  from fastapi import FastAPI
2
3
4  api = FastAPI()
5
6
7  @api.get("/")
8  def get_index(argument1):
9      return {
10         "data": argument1
11     }
12
13  @api.get("/typed")
14  def get_typed(argument1: int):
15      return {
16         "data": argument1 + 1
17     }
18
```

Run requests in terminal:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i http://127.0.0.1:8000/typed?argument1=1234
HTTP/1.1 200 OK
date: Thu, 12 Jun 2025 14:14:30 GMT
server: uvicorn
content-length: 13
content-type: application/json

{"data":1235}
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i http://127.0.0.1:8000/typed?argument1=hello
HTTP/1.1 422 Unprocessable Entity
date: Thu, 12 Jun 2025 14:14:38 GMT
server: uvicorn
content-length: 157
content-type: application/json

{"detail":[{"type":"int_parsing","loc":["query","argument1"],"msg":"Input should be a valid integer, unable to parse string as an integer","input":"hello"}]}
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$
```

Open URL <http://127.0.0.1:8000/typed?argument1=1234>

```
{
  "data": 1235
}
```

Open URL <http://127.0.0.1:8000/typed?argument1=hello>

```
{
  "detail": [
    {
      "type": "int_parsing",
      "loc": [
        "query",
        "argument1"
      ],
      "msg": "Input should be a valid integer, unable to parse string as an integer",
      "input": "hello"
    }
  ]
}
```

Open <http://127.0.0.1:8000/redoc>

The screenshot shows the Redoc API documentation interface. On the left is a sidebar with a search bar and two endpoints: 'Get Index' and 'Get Typed'. The main area displays the 'Get Index' endpoint details, including its query parameters (argument1, required, any (Argument1)) and response samples (200 Successful Response, 422 Validation Error). Below this, the 'Get Typed' endpoint is partially visible, showing its query parameters (argument1, required, integer (Argument1)). A dark-themed modal window is overlaid on the 'Get Typed' endpoint, showing response samples for status codes 200 and 422. The modal displays the content type 'application/json' and the response body 'null'.

Return to [main.py](#) and modify:

```
main.py 1, M X
datascientest-data-engineering > fast-api > main.py > get_ac
1  from fastapi import FastAPI    Import "fastapi" cou
2  from typing import Optional
3
4
5  api = FastAPI()
6
7  @api.get("/")
8  def get_index(argument1):
9      return {
10         "data": argument1
11     }
12
13  @api.get("/typed")
14  def get_typed(argument1: int):
15      return {
16         "data": argument1 + 1
17     }
18
19  @api.get("/addition")
20  def get_addition(a: int, b: Optional[int] = None):
21      if b:
22         result = a + b
23      else:
24         result = a + 1
25      return {
26         "addition_result": result
27     }
28
```

Open URL <http://127.0.0.1:8000/docs>

FastAPI 0.1.0 OAS 3.1
/openapi.json

default

GET / Get Index

GET /typed Get Typed

GET /addition Get Addition

Schemas

HTTPValidationError > Expand all object

ValidationError > Expand all object

Query for example, integers ten plus five:

GET /addition Get Addition

Cancel

Parameters

Name	Description
a ★ required	
integer (query)	10
b	
integer (integer null) (query)	5

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/addition?a=10&b=5' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/addition?a=10&b=5
```

Server response

Code	Details
200	Response body<pre>{ "addition_result": 15 }</pre> Download Response headers<pre>content-length: 22 content-type: application/json date: Thu, 12 Jun 2025 14:32:38 GMT server: uvicorn</pre>